

Freight Profit Analysis — Updated Pricing v4

Shipment: Matheus Reis (Ilhéus, Brazil) → Kirsten Ritschel (San Francisco, CA)

Date: Manifest 4/6/2026 · **Total Weight:** 375.08 kg · **Pallet:** 100×120×100 cm

Pricing Source: Gary Teh (governor) — updated 2026-06-11

SECTION 1: PAYLOAD & PRICING

1a. Pricing Structure

Product	Unit Size	Retail Price	Wholesale Price
Ceremonial Cacao Pouch	200g (7.05 oz)	\$25.00	\$17.00
Cacao Nibs Kraft Pouch	8 oz (227g)	\$25.00	\$17.00
Cacao Mass Bar (500g)	500g	\$50.00	\$34.00
Cacao Nibs (bulk, repacked)	227g pouch	\$25.00	\$17.00
Cacao Almonds (bulk, repacked)	200g pouch	\$25.00	\$17.00
Chocolate Bar (50g)	50g	\$10.00	\$6.00
Cacao Husk / Tea (to apothecaries)	Per lb	\$20.00/lb	\$20.00/lb
Cacao Tea (bulk, to apothecaries)	Per lb	\$20.00/lb	\$20.00/lb

1b. Line-by-Line Inventory & Revenue

#	Item	Qty	Unit Size	Total Wt (kg)	Channel	Price/unit	Revenue
1	8 oz Cacao Nibs Kraft Pouch [Main]	137	227g	31.07	Retail	\$25.00	\$3,425.00
2	Cacao Husk (KG) [Main] → apothecaries	20	1 kg = 2.2046 lb	20.00	Direct (\$20/lb)	\$44.09/kg	\$881.80
3	Cacao Mass Bar (500g) [Main]	38	500g	19.00	Retail	\$50.00	\$1,900.00

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#	Item	Qty	Unit Size	Total Wt (kg)	Channel	Price/unit	Revenue
4	Cacao Nibs (KG) [Main] → repack into 227g pouches	80	→ 352 pouches	80.00	Retail	\$25.00	\$8,800.00
5	Cacao Almonds (KG) [AGL8] → repack into 200g pouches	10	→ 50 pouches	10.00	Retail	\$25.00	\$1,250.00
6	Cacao Tea (KG) [AGL8] → apothecaries	12	1 kg = 2.2046 lb	0.01	Direct (\$20/lb)	\$44.09/kg	\$529.08
7	Ceremonial Cacao Pouch 200g (Paulo) [AGL8]	170	200g	34.00	Retail	\$25.00	\$4,250.00
8	Cacao Almonds KG (Vivi) [AGL13] → repack into 200g pouches	15	→ 75 pouches	15.00	Retail	\$25.00	\$1,875.00
9	Cacao Nibs (KG) Santos [AGL13] → repack into 227g pouches	100	→ 440 pouches	100.00	Retail	\$25.00	\$11,000.00
10	Cacao Tea (KG) Santos [AGL13] → apothecaries	21	1 kg = 2.2046 lb	21.00	Direct (\$20/lb)	\$44.09/kg	\$925.89
11	Cacao Almonds KG (Oscar) [AGL14] → repack into 200g pouches	10	→ 50 pouches	10.00	Retail	\$25.00	\$1,250.00

#	Item	Qty	Unit Size	Total Wt (kg)	Channel	Price/unit	Revenue
12	Pallet packaging	1	—	35.00	—	—	—
	TOTAL	613		375.08 kg			\$36,086.77

1c. Revenue Breakdown by Channel

Channel	Items	Revenue
Retail (ceremonial, nibs, mass bars, almonds)	1,312 units	\$32,750.00
Direct to apothecaries (husk + tea, 53 kg = 116.8 lb @ \$20/lb)	116.8 lb	\$2,336.77
Total Retail Value		\$36,086.77

SECTION 2: FREIGHT COST

2a. Current Estimated Cost

Cost Component	Amount (USD)
Air Freight (airport to airport) — 375 kg × \$3.40/kg	\$1,261.20
Export Documentation	\$95.00
Inland Transport (Brazil) — \$695 + 0.15% ad valorem	\$697.81
Brazil Airport Charges — max(\$0.30×375, \$250)	\$250.00
US Airline Terminal Fee (TAP)	\$212.50
US Import Handling Fee	\$125.00
US Customs Clearance	\$150.00
MPF (Merchandise Processing Fee)	\$33.58
Total Freight Cost	\$2,825.09

2b. Potential Cost Increases

Risk Factor	Original	Potential New	Delta
Inland Transport (Brazil)	\$697.81	~\$1,458.00 (R\$7,290 quoted)	+\$760.19
Pallet + Fumigation (new)	\$0	~\$139.00 (BRL 695)	+\$139.00

Risk Factor	Original	Potential New	Delta
Air Freight (war risk surcharge)	\$1,261.20	TBD	TBD
Worst-case freight total	\$2,825.09	~\$3,724+	+\$899+

2c. Freight Metrics

Metric	Current	Worst Case
Freight cost per kg	\$7.53/kg	~\$9.93/kg
Freight as % of retail value	7.8%	~10.3%
Freight cost per item (613 items)	\$4.61	~\$6.07

SECTION 3: COST OF GOODS SOLD

3a. Cacao Cost Basis (from Treasury Ledger)

Item	Qty	Unit Cost	Total COGS
8 oz Cacao Nibs Kraft Pouch [Main]	137	\$6.64	\$909.68
Cacao Husk (KG) [Main]	20	\$17.79/kg	\$355.71
Cacao Mass Bar (500g) [Main]	38	\$15.70	\$596.41
Cacao Nibs (KG) [Main]	80	\$24.62/kg	\$1,969.48
Cacao Almonds (KG) [AGL8]	10	\$0.00	\$0.00
Cacao Tea (KG) [AGL8]	12	\$0.00	\$0.00
Ceremonial Cacao Pouch 200g (Paulo) [AGL8]	170	\$10.37	\$1,762.90
Cacao Almonds KG (Vivi) [AGL13]	15	\$7.87/kg	\$118.05
Cacao Nibs (KG) Santos [AGL13]	100	\$10.18/kg	\$1,018.23
Cacao Tea (KG) Santos [AGL13]	21	\$10.18/kg	\$213.83
Cacao Almonds KG (Oscar) [AGL14]	10	\$11.41/kg	\$114.12
Total Cacao COGS			\$7,058.41

3b. Packaging Costs (for bulk items needing repacking)

Bulk items to repack:

- Cacao Nibs (KG) [Main]: 80 kg → 352 × 227g pouches
- Cacao Nibs (KG) Santos [AGL13]: 100 kg → 440 × 227g pouches
- Cacao Almonds (KG) [AGL8]: 10 kg → 50 × 200g pouches
- Cacao Almonds KG (Vivi) [AGL13]: 15 kg → 75 × 200g pouches
- Cacao Almonds KG (Oscar) [AGL14]: 10 kg → 50 × 200g pouches

Item	Qty	Unit Cost	Total
Kraft pouches for nibs (792 pouches)	792	\$0.70	\$554.40
Kraft pouches for almonds (175 pouches)	175	\$0.74	\$129.50
Labels (Sticker Mule)	967	\$0.55	\$531.85
Shipping boxes	20	\$0.55	\$11.00
Total Packaging			\$1,226.75

3c. Total COGS

Component	Amount
Cacao cost basis	\$7,058.41
Packaging materials	\$1,226.75
Total COGS	\$8,285.16

SECTION 4: REVENUE SCENARIOS

4a. All Retail (100% at retail prices)

Channel	Calculation	Revenue
137 × Nibs pouches @ \$25	137 × \$25	\$3,425.00
20 kg Husk @ \$44.09/kg (apothecaries)	20 × \$44.09	\$881.80
38 × Mass Bars @ \$50	38 × \$50	\$1,900.00
352 × Nibs pouches (from 80 kg bulk) @ \$25	352 × \$25	\$8,800.00

Channel	Calculation	Revenue
50 × Almond pouches (from 10 kg) @ \$25	50 × \$25	\$1,250.00
12 kg Tea @ \$44.09/kg (apothecaries)	12 × \$44.09	\$529.08
170 × Ceremonial pouches @ \$25	170 × \$25	\$4,250.00
75 × Almond pouches (from 15 kg) @ \$25	75 × \$25	\$1,875.00
440 × Nibs pouches (from 100 kg) @ \$25	440 × \$25	\$11,000.00
21 kg Tea @ \$44.09/kg (apothecaries)	21 × \$44.09	\$925.89
50 × Almond pouches (from 10 kg) @ \$25	50 × \$25	\$1,250.00
Total Revenue (All Retail)		\$36,086.77

4b. All Wholesale (100% at wholesale prices)

Channel	Calculation	Revenue
137 × Nibs pouches @ \$17	137 × \$17	\$2,329.00
20 kg Husk @ \$44.09/kg (apothecaries — fixed)	20 × \$44.09	\$881.80
38 × Mass Bars @ \$34	38 × \$34	\$1,292.00
352 × Nibs pouches @ \$17	352 × \$17	\$5,984.00
50 × Almond pouches @ \$17	50 × \$17	\$850.00
12 kg Tea @ \$44.09/kg (apothecaries — fixed)	12 × \$44.09	\$529.08
170 × Ceremonial pouches @ \$17	170 × \$17	\$2,890.00
75 × Almond pouches @ \$17	75 × \$17	\$1,275.00
440 × Nibs pouches @ \$17	440 × \$17	\$7,480.00
21 kg Tea @ \$44.09/kg (apothecaries — fixed)	21 × \$44.09	\$925.89
50 × Almond pouches @ \$17	50 × \$17	\$850.00
Total Revenue (All Wholesale)		\$25,286.77

4c. Mixed Scenario (30% Retail / 70% Wholesale)

Retail items (excluding apothecary fixed-price items):

- Retail value: \$32,750.00

- Wholesale value: \$22,400.00 (using \$17/\$34 wholesale prices)

Channel	Calculation	Revenue
Retail portion (30% of retail items)	$\$32,750 \times 30\%$	\$9,825.00
Wholesale portion (70% at wholesale prices)	$\$22,400 \times 70\%$	\$15,680.00
Apothecaries (husk + tea — fixed price)		\$2,336.77
Total Revenue (Mixed)		\$27,841.77

4d. Aora/Go China — 100 Chocolate Bars

Metric	Value
Bars from this shipment (Kirsten produces)	63 new + 37 already sold = 100 total
DAO return per bar	\$6.00
Total DAO revenue from Aora	\$600.00
Production cost (~\$2.50/bar)	\$250.00
Net profit from Aora bars	\$350.00

SECTION 5: PROFIT & LOSS STATEMENTS

5a. Scenario A — All Retail

P&L Line	Amount
Revenue	\$36,086.77
Less: Freight Cost	-\$2,825.09
Less: Cacao COGS	-\$7,058.41
Less: Packaging Materials	-\$1,226.75
Less: Payment Processing (Stripe ~2.9% + \$0.30)	-\$1,080.00
Less: Platform Fees	-\$100.00
Gross Profit	\$23,796.52
Gross Margin	65.9%
Return on Freight Investment	8.4x

5b. Scenario B — All Wholesale

P&L Line	Amount
Revenue	\$25,286.77
Less: Freight Cost	-\$2,825.09
Less: Cacao COGS	-\$7,058.41
Less: Packaging Materials	-\$1,226.75
Less: Payment Processing	-\$760.00
Gross Profit	\$13,416.52
Gross Margin	53.1%
Return on Freight Investment	4.7×

5c. Scenario C — Mixed (30% Retail / 70% Wholesale)

P&L Line	Amount
Revenue	\$27,841.77
Less: Freight Cost	-\$2,825.09
Less: Cacao COGS	-\$7,058.41
Less: Packaging Materials	-\$1,226.75
Less: Payment Processing	-\$835.00
Gross Profit	\$15,896.52
Gross Margin	57.1%
Return on Freight Investment	5.6×

5d. Scenario D — Worst Case (Increased Freight + Wholesale)

P&L Line	Amount
Revenue	\$25,286.77
Less: Freight Cost (worst case ~\$3,724)	-\$3,724.00
Less: Cacao COGS	-\$7,058.41
Less: Packaging Materials	-\$1,226.75
Gross Profit	\$13,277.61
Gross Margin	52.5%
Return on Freight Investment	3.6×

SECTION 6: UNIT ECONOMICS

6a. Per-Kilogram Economics

Metric	Retail Scenario	Wholesale Scenario
Revenue per kg shipped	\$96.23/kg	\$67.43/kg
Freight cost per kg	\$7.53/kg	\$7.53/kg
COGS per kg	\$22.09/kg	\$22.09/kg
Profit per kg	\$63.46/kg	\$35.77/kg

6b. Per-Item Economics (1,312 sellable units)

Metric	Retail	Wholesale
Avg revenue per item	\$27.50	\$19.27
Avg cost per item (freight + COGS)	\$8.47	\$8.47
Avg profit per item	\$19.03	\$10.80

6c. Break-Even Analysis

Metric	Value
Break-even revenue (current freight)	\$11,110.25
Break-even as % of retail value	30.8%
Units to sell at retail to break even	~404 units
Units to sell at wholesale to break even	~577 units

SECTION 7: LA DO SITIO FULL CONVERSION (274 kg — Future)

Conversion Math

Step	Value
Cacao almonds available (AGL8)	274 kg
Cacao content per 50g bar (81%)	40.5g
Theoretical max bars	~6,765 bars
Realistic yield (~15% loss)	~5,750 bars
Pilot-scale first batch	~2,000 bars

Economics at Scale

Metric	Per Bar	2,000 Bars	5,750 Bars
Wholesale price (DAO return)	\$6.00	\$12,000.00	\$34,500.00
Production cost (Santos R\$130/kg)	\$1.30	\$2,600.00	\$7,475.00
Freight to China (100 kg @ \$3.40/kg)	\$0.17	\$340.00	\$977.50
Net Profit	\$4.53	\$9,060.00	\$26,047.50
Margin	75.5%	75.5%	75.5%

SECTION 8: CASH FLOW & CAPITAL POSITION

8a. Current Deployable Cash

Ledger	Amount
Main Ledger (Gary)	\$3,172.29
AGL15 (Gary)	\$5,279.73
Total	\$8,452.02
Less: Current Freight	-\$2,825.09
Remaining after freight	\$5,626.93

8b. Cash Required to Sell the Shipment

Cost	Amount	Covered By
Freight (already allocated)	\$2,825.09	Main + AGL15
Packaging materials	\$1,226.75	Remaining cash
Kirsten's production labor (63 bars)	~\$200.00	Remaining cash
Total cash needed	\$4,251.84	
Cash remaining after all costs	\$1,375.09	

8c. ROI on Cash Deployed

Metric	Value
Total cash deployed (freight + packaging + labor)	\$4,251.84
Expected return (mixed scenario)	\$15,896.52
Cash-on-cash return	3.7×
Net profit on cash deployed	\$11,644.68

SECTION 9: KEY RISKS & SENSITIVITY

9a. Risk Factors

Risk	Impact	Probability	Mitigation
Air freight rate increase (war risk)	+\$200–\$500	Medium	Lock rate ASAP
Inland transport increase (confirmed)	+\$760	High	Already quoted R\$7,290
Customs exam fee	+\$250–\$500	Low	Proper documentation
Damage/loss in transit	Full loss	Very Low	Insurance
Slow sell-through (cash tied up)	Delayed ROI	Medium	Wholesale channels
Kirsten production delay	Delayed Aora bars	Medium	Parallel Santos production

9b. Sensitivity Table — Profit vs. Sell-Through Rate (Mixed Scenario)

Sell-Through	Revenue	Profit	Margin
25%	\$6,960	-\$3,150	-45%
50%	\$13,921	\$3,811	27%
75%	\$20,881	\$10,771	52%
100%	\$27,842	\$15,897	57%

9c. Time to Profitability

Assuming \$1,000/month in retail sales (current run rate):

- Break-even at mixed pricing: ~7 months
- Break-even at wholesale (bulk sale): Immediate

SECTION 10: SUMMARY

Scenario	Revenue	Total Costs	Net Profit	Margin	ROI
All Retail	\$36,086.77	\$12,290.25	\$23,796.52	65.9%	8.4×
All Wholesale	\$25,286.77	\$11,870.25	\$13,416.52	53.1%	4.7×
Mixed (30/70)	\$27,841.77	\$11,945.25	\$15,896.52	57.1%	5.6×
Worst Case	\$25,286.77	\$12,749.16	\$12,537.61	49.6%	3.6×
Aora 100 bars	\$600.00	\$250.00	\$350.00	58.3%	1.4×
La do Sitio (2K bars)	\$12,000.00	\$2,940.00	\$9,060.00	75.5%	3.1×

Key Takeaways

1. **\$50/bar for 500g mass bars adds \$950 extra revenue** at retail vs standard pricing
2. **Apothecaries at \$20/lb adds \$2,337** for husk and tea — excellent channel
3. **Even at wholesale, profit is \$13,417** — 53% margin is very strong
4. **Freight cost is only 7.8% of retail value** — outstanding ratio
5. **Cash-on-cash return of 3.7×** in the mixed scenario
6. **Biggest risk is sell-through speed**, not freight cost